

OSF Protocol

1. Title

Sustainable Building Materials in Africa (2015–2025): A Systematic Review of Material Types, Performance Metrics, and Implementation Contexts

2. Contributors

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Roles: Conceptualization, literature search, screening, data extraction, quality appraisal, synthesis, manuscript preparation

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3. Background and Rationale

Rapid urbanisation across Africa has intensified demand for affordable, durable, and environmentally responsible building solutions. Sustainable building materials (SBMs) including earth-based, bio-based, recycled, and low-carbon binders offer potential to reduce environmental impact, enhance affordability, and support local economies. However, the evidence base is fragmented across countries, disciplines, and study designs, limiting its utility for policy and practice. This systematic review will consolidate and critically appraise research published between **1 January 2015 and 31 August 2025**, providing a coherent synthesis to guide decision-makers, practitioners, and researchers.

Operational definition: For this review, SBMs are defined as materials meeting at least one of the following:

- Low embodied carbon or energy
- Renewable or bio-based source

- Recycled content
- Improved durability or performance with reduced environmental impact

4. Objectives

Primary Objective: Identify and synthesise evidence on SBMs used or studied in African contexts between 2015–2025, focusing on material types, performance characteristics, and environmental, economic, technical, and social/practical impacts.

Secondary Objective: Identify research gaps, barriers, and enabling factors influencing SBM adoption.

Tertiary Objective: Map geographic and climatic contexts of SBM studies in Africa.

5. Review Question

What sustainable building materials have been investigated or applied in African contexts from 2015–2025, and what are their environmental, economic, technical, and social/practical performance outcomes?

6. Eligibility Criteria

Population/Context:

- Studies conducted in African countries or explicitly addressing African contexts.

Focus:

- Use, testing, or assessment of sustainable/low-impact building materials (e.g., stabilised earth, bamboo, recycled aggregates, geopolymers, cement, calcined clay).

Comparators:

- Conventional materials or alternative SBMs (where applicable).

Outcomes:

- **Environmental:** embodied carbon, energy use, LCA indicators
- **Economic:** cost, life-cycle cost, affordability
- **Technical:** strength, durability, thermal/acoustic performance
- **Social/practical:** local availability, skills, codes/standards, user acceptance

Study Types:

- Peer-reviewed journal articles, conference papers, theses with clear methods, case studies, LCA studies, experimental lab/field studies.

Timeframe:

- **1 January 2015 – 31 August 2025**

Language:

- English; non-English studies included if an English abstract contains methods and at least one extractable outcome measure.

Exclusions:

- Opinion pieces, patents, news/blogs, studies outside Africa with no African relevance.

7. Information Sources

- **Databases:** Google Scholar, African Journals Online (AJOL), Directory of Open Access Journals (DOAJ), Scopus (if accessible), Web of Science (if accessible)
- **Grey Literature:** Government/NGO reports with explicit methodology, conference proceedings
- **Supplementary:** Backward and forward citation tracking of included studies
- **Note:** ResearchGate used only to access author-uploaded full texts, not as a primary search engine.

8. Search Strategy

Example Google Scholar string:

- ("sustainable building materials" OR "green construction" OR "low carbon" OR "life cycle assessment" OR "LCA")
- AND (Africa OR Nigeria OR Kenya OR Ghana OR "South Africa" OR Ethiopia OR Rwanda OR Egypt OR Morocco)
- AND (2015..2025)

Additional material-specific terms: “*compressed earth blocks*”, “*stabilised soil*”, “*laterite*”, “*bamboo*”, “*geopolymer*”, “*recycled aggregate*”, “*calcined clay*”, “*LC3*”. Search strings will be adapted for each database. All search strings, dates, and filters will be documented in the Search Logs component.

9. Study Records and Data Management

- **Reference Management:** Zotero (free)
- **Deduplication:** Zotero duplicate detection
- **Screening:** Dual-reviewer title/abstract screening, then full-text screening; disagreements resolved by discussion or third reviewer.
- **Storage:** PDFs, extraction sheets, and appraisal forms stored in OSF project files with version control.

10. Selection Process

- **Stage 1:** Title/abstract screening against inclusion criteria
- **Stage 2:** Full-text screening with detailed exclusion reasons
- **Documentation:** PRISMA 2020 flow diagram to illustrate selection

11. Data Extraction

Tool: Excel or Google Sheets

Fields: Author, Year, Journal/Source, Country, Region, Climate Context, Material Type, Composition, Study Design, Scale, Environmental Outcomes, Economic Outcomes, Technical Outcomes, Social/Practical Notes, Comparator, Standards Referenced, Key Findings, Limitations, Funding/Conflicts, Quality Tool Used, Risk of Bias, Appraisal Notes.

12. Risk of Bias / Quality Assessment

- **Quantitative/experimental/observational/economic studies:** JBI Critical Appraisal Checklists
- **Qualitative/mixed-methods studies:** CASP Qualitative Checklist
- **LCA studies:** ISO 14040/44 alignment noted

- Disagreements resolved by discussion or third reviewer.
- All appraisal decisions documented in the extraction sheet.

13. Data Synthesis

- **Approach:** Narrative synthesis organised by material category and outcome domain
- **Tabulation:** Summary tables by material and country; medians/ranges for comparable quantitative metrics
- **Subgrouping:** By region, scale, and material class
- **Heterogeneity:** Described qualitatively; no meta-analysis planned due to methodological diversity

14. Reporting

- **Guideline:** PRISMA 2020
- **Outputs:** PRISMA checklist, flow diagram, summary tables, narrative synthesis
- **Availability:** All outputs uploaded to OSF upon manuscript submission

15. Timeline

(Rapid review)

- **Week 1:** Finalise protocol, pilot searches, register on OSF
- **Week 2:** Full searches, title/abstract screening
- **Week 3:** Full-text screening, data extraction, quality appraisal
- **Week 4:** Synthesis, write-up, submission to target journal

16. Ethics and Dissemination

- **Ethics:** Not required (secondary analysis of published data)
- **Dissemination:** Submission to peer-reviewed journal, OSF preprint, plain-language summary for practitioners and policymakers
- **Data Availability Statement:** All data supporting the findings of this review, including the registered protocol, search strategies, screening decisions, data

extraction sheets, and quality appraisal summaries, will be available in the OSF project upon manuscript submission. Final search date: 31 August 2025.